

CMPS310 Software Engineering
- Group Project -
Milestone-2: Design and Implementation (13%)

This is a compulsory project. If you do not submit this, your grade will be 0 in this course.

Submission Due: 16/11/2024 (mid-night) on Blackboard

Fall 2024

Mr. Khalid Hassan from the Qatar Vehicle Registration (QVR) department has reviewed your *Milestone-1: Requirements Analysis and Design*. He has decided that your team should continue developing the project further. This time, he requests that your team create sequence diagrams for two use cases, identify important system properties (constraints and quality requirements), implement two use cases, and test the program. This document contains the following items:

- (1) Your tasks, grade distribution, and submission requirements
- (2) The same system description as in Milestone-1, available in **Appendix A**
- (3) Two use case descriptions, provided in **Appendix B**

Your Tasks: This project concentrates on the technical aspects of the software engineering process, with an emphasis on quality properties and architectural design issues. The body of your Milestone 2: Design and Implementation will consist of the following deliverables:

- 1) Prepare two sequence diagrams for any two use cases proposed in your Milestone-1 submission. You must also include the use case specifications with the normal scenarios for those two use cases, as well as a class diagram, in your Milestone-2 submission. However, you may update or improve them if desired (20%).
- 2) Identify Constraints and categorize them into different groups. Also, recognize the quality requirements/Non-Functional Requirements (NFRs) of the system. You can propose additional constraints and NFRs beyond the system requirements. Most of the necessary information is available in the systems requirements in **Appendix-A** (20%). **Note:** Desired quality attributes should be specified precisely with exact scenarios. For instance, instead of simply stating that the system is scalable, you should be more specific such as "The system can handle more than 50 thousands orders within next 3 years."
- 3) Implement the two use cases presented in **Appendix-B** using Java and provide screenshots demonstrating that your program works, along with some sample output. Use a well-designed user interface to give your program a professional look. You must provide the normal scenario of the use case (20%).
- 4) Test the program developed in Task 3. You must present the test cases and their results, and explain how the program was tested. (20%)
- 5) Test at least two quality requirements (NFRs) identified in Task (2). You must demonstrate how you tested them, and explain the test results. **Note:** You may use any testing technique(s) discussed during the lectures to complete this task (20%).

Provide justifications for any assumptions you made in this milestone. Your submitted components will be evaluated for accuracy, clarity, relevance, justification, and completeness (especially among components and artifacts) of your document.

Grading scheme for Milestone-2: Requirements Analysis

Deliverables	Grading %
Task (1) Two sequence diagrams Submit <i>normal scenarios of two use cases and design class diagram</i>	20
Task (2) Constraints and quality properties	20
Task (3) Implementation of one use case with nice interface	20
Task (4) Testing the program developed in Task (3)	20
Task (5) Testing two NFRs	20
Total	100

Submission requirements

Your assignment group/team **MUST** comply with the following submission requirements; **otherwise grade will be deducted:**

1. **Make sure that each team member must submit the identical electronic copy individually on Blackboard.**
2. Add a snapshot from your Github page and make sure that you include the link of Github page.
3. The cover page of your submission MUST contain the following items in four bullets:

[Each member of the group will be required to submit a peer group evaluation with an estimate of the contribution from each member to the project. Write the percentage contribution made by each team member so that it adds up to 100%. This evaluation *may* be used to adjust the marks awarded to each team member]

- Effort distribution of the student:

QUID: _____	STUDENT NAME: _____	Effort given _____	%
QUID: _____	STUDENT NAME: _____	Effort given _____	%
QUID: _____	STUDENT NAME: _____	Effort given _____	%
QUID: _____	STUDENT NAME: _____	Effort given _____	%

- Class section: **L02 / L03 / L51 / L52** (select one)

- Course number _____

- Submission date _____

- DECLARATION: *We hereby certify that no part of this project or product has been copied from any other student's work or from any other sources except where due acknowledgement is made in the project. No part of this project/product has been written/produced for us by any other persons.*

4. Be aware of

- *Submitted work must be students' own work*
- *You cannot copy the project from other groups.*

Appendix-A: Qatar Vehicle Registration System

The systems requirements are the same as in Milestone-1, but the additional information on page 4 is new.

Systems Requirements

Assume that Mr. Khalid Hassan is the newly appointed Director of the Qatar Vehicle Registration (QVR) department. Currently, the department offers various services such as vehicle registration renewal, ownership transfer, management of compulsory insurance policies, fitness certificates, accident reporting, and penalty handling. These services are not yet available online to the general public. Mr. Khalid has decided to make these key services available online through a platform called iQVR for the public in Qatar. Recently, Mr. Khalid contacted your group to develop the online system for the iQVR department. He has provided a description of the major functions expected to be available online and has emphasized that the scope of your analysis, design tasks, and system boundary relates completely to QVR.

Currently, iQVR comprises five sections: The Finance section handles payments, the Penalty section manages fines and offences, the Registration section processes renewals and ownership transfers, the Accident section addresses all accident-related matters, and the Technical section manages vehicle fitness. The department uses a computer-based system for managing all payments and employee salaries, which is a standard commercial-off-the-shelf package known as qPay. Mr. Khalid now intends to replace all five sections with an automated online system while keeping qPay as a separate, connected system. He plans to employ only one system administrator to manage this automated process. iQVR should function as described below.

Vehicle owners, insurance companies, and authorized workshops will need a login ID and password to access the iQVR online service. It is assumed that users are already logged in by default, so login functionality does not need to be designed or developed by your team.

To register a new vehicle or renew its registration, the vehicle owner must first purchase an insurance policy from an insurance company. Once the policy is bought, it is immediately recorded in the iQVR system. Each vehicle must have its own insurance policy, and a policy is valid for only one vehicle. The company enters the Vehicle Identification Number (VIN) into the system and attaches a copy of the policy. The system then records this information along with the company data and generates an acknowledgment receipt for the company.

If the vehicle is over two years old, it requires a fitness certificate from an authorized workshop in Qatar. Once the vehicle obtains a fitness certificate, the workshop submits it online to the iQVR system by entering the VIN and certificate number and attaching a copy of the certificate. Each certificate is valid for only one vehicle; however, a workshop can issue multiple certificates. The system stores this information along with the workshop data and generates an acknowledgment receipt for the workshop. A registered vehicle can have only one current owner, but an owner can have multiple vehicles.

To transfer the ownership of a new, unregistered vehicle (one that has not been registered before) to another person, the current owner enters the VIN, vehicle make, model, and year of manufacture into the system. The system then provides the VIN and car details to another system operated by the vehicle manufacturer. This system verifies the VIN and other specifications with its factory database. If the manufacturer confirms the information, the iQVR system requests the current owner's name and QID. If the manufacturer's response is negative, the system terminates the session with the message "Incorrect vehicle information". Next, the iQVR system communicates with another system run by Qatar Trade Service, using the VIN and QID of the current owner, to check if the owner imported the vehicle from overseas. Qatar Trade Service checks its records and provides a response to iQVR. If the response is positive, the system then retrieves the insurance policy for the vehicle, which was previously submitted by the insurance company. If the response is negative, the system terminates the session with the message "Incorrect ownership." If the insurance policy exists, the system generates a registration number for the vehicle, assigns it to the vehicle, and displays the number to the owner. The current owner then provides the name, QID, address, and mobile phone number of the new owner. The system records these details and assigns the vehicle to the new owner. It designates the current owner as the 'previous owner' and the new

owner as the 'current owner'. The system then confirms the ownership transfer by generating a new registration sticker and creating an invoice for the transfer fee, which the new owner will pay later.

However, to transfer a registered vehicle to another person, the current owner enters the VIN of the registered vehicle and the QID number of the current owner. The system first retrieves the registration details of the vehicle. If the retrieved information does not match the provided data, the system terminates the session with the message "Incorrect Information". Otherwise, it initiates the transfer process. The system checks for any unpaid invoices (bills) for registration fees or traffic fines. If there are any unpaid bills, the system terminates the session with the message "Pay the bills first". The payment procedure is explained later. If there are no unpaid dues, the system requests the new owner's details. The current owner provides the name, QID, and mobile phone number of the new owner. The system then assigns the vehicle to the new owner, designates the current owner as the 'previous owner' and the new owner as the 'current owner'. The insurance policy assigned to the vehicle during the last registration remains unchanged; however, the new owner can update the insurance policy later. The system confirms the ownership transfer by generating a new registration sticker and creating an invoice for the transfer fee.

To renew a registered vehicle, the vehicle owner enters the VIN into the system. It retrieves the registration details and checks if the vehicle is over two years old. If it is, the system finds the vehicle fitness certificate issued by an authorized workshop. The system terminates with a message "Get fitness certificate first" if the fitness certificate is not available. If the certificate exists, it retrieves the compulsory insurance policy of the vehicle issued by an insurance company. If a new insurance policy is available, the system checks for any unpaid fines for traffic offences; otherwise, terminate the system with the message "Get insurance policy". If there is any unpaid fine(s), it asks the owner to pay the fine(s) with the message "Pay the bill first". If there are no unpaid fines (s), the system creates a new registration with the same validation period as specified in the new insurance policy. The insurance policy and the fitness certificate are then attached concurrently with the vehicle registration. The system then prepares a new registration sticker with new validity. It finally creates an invoice for the registration renewal.

For an accident between only two vehicles, if a vehicle accepts that it was his/her fault, he/she enters VINs of his/her and the other vehicle (victim). The system finds vehicle details and registration information for both vehicles. The system then asks to provide the date, time, location, and a brief accident description. The system records this information, and by default, it sets that the vehicle that entered all information is the offending vehicle's owner. The other vehicle is the victim of the accident. The system asks the owner of the offending vehicle to confirm this. Once confirmed, the system stores this. The system then concurrently finds the insurance policy of the offending vehicle and creates an accident report with a unique case number. The iQVR system then sends the accident report to the insurance company, which immediately sends an acknowledgement receipt saved by the iQVR system. The accident report is then stored and available for both parties (owners of the offending and victim vehicles) to retrieve at any time. Any authorized workshop can retrieve the accident report if the case number is known. However, if there is a dispute regarding an accident, no online reporting is allowed. Both parties must go to a police station to solve the dispute.

Vehicle owners can pay any unpaid invoices or fines for a vehicle using a credit card. To pay, the owner provides the VIN, and credit card details such as number, name of the cardholder, and validity. The system retrieves the vehicle registration details. If the registration details do not exist, the system displays an error message and asks to enter the correct VIN. Otherwise, it lists unpaid invoice(s), if there are any. The owner then selects which of the invoices and fines he/she wants to pay. The system saves the selection, computes the total amount, and then forwards the card details and the total amount to the qPay system for approval. qPay is a separate software system but owned by the OVR department. The qPay checks for the card's validity by contacting the credit card provider (bank). If the card is invalid, it generates an error message to iQVR without payment processing. Otherwise, qPay returns an approval advice to iQVR. For invalid credit cards, iQVR asks the owner to enter valid credit card details. The iQVR system records the outcome sent by qPay. It then concurrently records the approval number, prepares a payment receipt, and sets the invoice(s) as paid. It then displays the receipt. Finally, the system updates the list of the unpaid invoices associated with the vehicle.

The traffic police can determine which vehicle has how many red-light offences in the last year, one month or one week. He/she enters the period (day, week, month) and selects the red-light offence type from the system. Based on these criteria, the system finds a list of vehicles that match these criteria. The traffic police select those vehicles that exceed a certain number of red-light offences during this period. The system creates a confiscating order for those vehicles and asks for confirmation from the traffic police. It saves the confirmation, cancels the registration of those vehicles, and informs the owner of each vehicle. The owner receives the order. The system then broadcasts the orders to all police departments. A confiscating order is attached with only one vehicle; however, one can get more than one order.

The system knows the registration details of vehicles, models and the make of each vehicle; previous and current owners of each vehicle such as their name, QID, and address. It also knows the insurance policy of every registered vehicle, such as policy number, validity, and the name of the contact details of the issuing company of the policy. A company can sell many insurance policies. The system also stores information about the fitness certificate of each registered vehicle and the details of the workshop that issued the certificate. It keeps information about the invoices, fines, traffic offences and the involved vehicles. The system knows the details of each accident, such as the date, time of the accident, and the involved vehicles. However, it does not record any information about the traffic police except their login data. The system keeps records of all acknowledgements and responses that it receives from external systems.

Currently, the proposed system can handle 10 million vehicles. However, it is projected that the traffic system will have more than 30 million vehicles within the next 10 years. More new functionalities must be added to the system; some existing ones might be modified or deleted to enhance further capabilities. The core data about the vehicles and the ownership details are confidential; these should not have direct exposure to any general user-level functions. These need to be protected by separating them from the high-level user interface. The software could be portable, which means it can run on various machines such as mobile phones, tab, desktops, etc. From time to time, some functions may need upgrading and modifications, but the interfaces to the rest of the system may remain the same. The software could be distributed in future; different components may run on different machines in different locations. Several backup modules will be managed to support the availability of the systems to the users most of the time. Various software components should be less dependent on each other and more focused.

The traffic police department does not have the budget for more than 10 new technical staff for the project. It has a budget for only 20 new servers. The system must run as a plug-in for the registration system that uses the Oracle database system for data management. Most functions were developed using Java and C. The traffic police department has qualified staff experienced with Oracle, Java and C. The first version of the system should be completed within three months. Any delay will cost money for the traffic department because the system is expected to start operating precisely after 12 months.

Appendix-B

Two use cases that need to be implemented.

Use case Name:	Transfer registered vehicle
Brief Description:	This use case begins when the owner of a vehicle wants to transfer a registered vehicle. The system transfers the ownership to a new owner.
Primary actors:	Vehicle owner
Trigger: The owner of a vehicle enters the VIN of a registered vehicle and the owner’s details to transfer it to another owner.	
Preconditions: 1. The registration already exists	
Post-conditions: 1. The ownership was changed. 2. A new registration sticker was created 3. An invoice was created.	
Main Success Scenario: The ownership was transferred.	
Normal scenario: (Actor Action, System Response Table)	
Actor Action	System Response
1. Vehicle owner enters VIN, QID, name	2. Retrieve the registration details
	3. Verify the correctness of information (see alternative flow 3a. if the information is incorrect)
	4. Check for the unpaid bills (For unpaid bills, see alternative flow 4a.)
	5. Ask for detail of the new owner
6. Vehicle owner provides details of the new owner.	7. Assign details of new owner to the vehicle
	8. Set current and previous ownership.
	9. Prepare registration sticker.
	10. Create invoice.
Alternative flows (if any): 3a. If the retrieved information does not match, terminates the session with the message “Incorrect Information.” 4a. If there is any unpaid bill(s), the system terminates with the message “Pay the bills first.”	

Use case Name:	Manage accident
Brief Description:	This use case begins when the owner of a vehicle wants to report an undisputed accident that was due to his/her fault. The system handles the accident matter by sorting out the insurance claim issues.
Primary actors:	Vehicle owner, Insurance company
Trigger: The owner of a vehicle enters the VINs of the involved vehicles	
Preconditions: 1. The registration already exists of both vehicles 2. Only two vehicles are involved in the accident.	
Post-conditions: 1. The insurance company of the offending vehicle was informed with the accident details.	
Main Success Scenario: The insurance issue of the accident was settled.	
Normal scenario: (Actor Action, System Response Table)	
<u>Actor Action</u>	<u>System Response</u>
1. Vehicle owner enters VIN of two vehicles involved in an accident	2. Retrieve details of both vehicles
3. Vehicle owner inputs accident details	4. Record accident information
	5. Set the vehicle of the owner as the offending vehicle
	6. Set the other vehicle as victim
	7. Ask to confirm
8. Vehicle owner confirms these	9. Store the confirmation
	10. Find the insurance policies of the vehicle, and create an accident report
	11. Send the accident report to the insurance company
12. Insurance company sends acknowledgement	13. Save the acknowledgement
	14. Store the accident report
Alternative flows (if any):	